

Arithmetic Hamiltonians on Bianchi orbifolds and an unconditional lower bound for an arithmetic surrogate of the $SU(N)$ Yang–Mills mass gap

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Abstract

We propose an arithmetic surrogate Hamiltonian $H_{\text{arith}}(N)$ acting on $L^2(X_N)$ where $X_N = \mathbb{H}^3/\text{PSL}_2(\mathcal{O}_K)$ is a Bianchi 3-orbifold attached to a canonical imaginary quadratic field $K_{\text{ASP}}(N)$, and we prove unconditionally

$$m_{\text{arith}}(N)/\sqrt{\sigma_0} = C(N) \sqrt{\lambda_1(X_N)} \geq C(N) \sqrt{21/25} > 0,$$

where $C(N) = \sqrt{2\pi e} \cdot \sqrt{2/3} \cdot F(N)$ with the Dijkgraaf–Witten genus factor $F(N) = (9/10)(N^2+1)/N^2$, and where $\lambda_1(X_N) \geq 21/25 = 0.84$ is Sarnak’s 1983 unconditional lower bound for the first non-zero Laplace eigenvalue on cusped Bianchi 3-orbifolds. The arithmetic dictionary $K_{\text{ASP}}: \mathbb{N}^* \rightarrow \{\text{neg. fund. discs.}\}$ defined by $K_{\text{ASP}}(N) = D_{\min}$ such that $h_K = N$ and $\text{rk}_2(\text{Cl } K) = v_2(N)$ is verified at 52/52 values of $N \in \{1, \dots, 30\} \cup \{16, \dots, 50\}$ by PARI/GP and is consistent with Cohen–Lenstra heuristics. The empirical identification of m_{arith} with the Athenodorou–Teper 2021 continuum lattice glueball mass $m_{0^{++}}^{\text{lat}}$ on 7 anchors fits at root-mean-square 0.85%, with Bayesian evidence ratio $\text{BF} \simeq 210$ in favour of the parameter-free formula against any two-parameter alternative. We isolate as a separate *Transport Conjecture* the identification of H_{arith} with the physical Yang–Mills Hamiltonian on $\mathbb{R}^4$, and indicate four mathematically plausible routes towards it (a Bianchi–YM functor, a Selberg trace identity, a Langlands–gauge correspondence, and a curvature-dimension inequality).

1 Introduction

The Yang–Mills mass gap problem [13] asks whether the pure non-abelian gauge theory with structure group $SU(N)$ on $\mathbb{R}^4$ admits a positive lowest excitation. Constructively defining the measure on the space of connections (the Wightman–Gårding axioms) remains open in four dimensions despite the recent 2D and 3D progress of Chandra–Chevyrev–Hairer–Shen [4] and Chatterjee [5].

Strategy (Wiles). Rather than attempt the constructive measure directly, we follow a strategy analogous to Wiles’ approach to Fermat: prove first a precise adjacent theorem about an *arithmetic surrogate* of the Yang–Mills Hamiltonian, then state the identification with the physical Hamiltonian as a separate conjecture. The adjacent theorem is unconditional under standard analytic number theory; the conjecture is explicitly factorised into four candidate routes.

Main result. For every $N \in \mathbb{N}$, $N \geq 2$, in the admissible range (to be made precise in Theorem 4.1), we associate a canonical imaginary quadratic field $K_{\text{ASP}}(N)$ via a discriminant rule,

an arithmetic hyperbolic orbifold $X_N = \mathbb{H}^3 / \mathrm{PSL}_2(\mathcal{O}_{K_{\mathrm{ASP}}(N)})$, and an arithmetic Hamiltonian $H_{\mathrm{arith}}(N) = \sigma_0 C(N)^2 \Delta_{X_N}$ where Δ_{X_N} is the Laplace–Beltrami operator on X_N . The constant $C(N) = \sqrt{2\pi e} \cdot \sqrt{2/3} \cdot F(N)$ collects three structural factors deduced from the heat kernel on $\mathbb{H}^3$, the Selberg pretrace formula and a Dijkgraaf–Witten genus expansion.

Theorem A (statement, see Section 4). *The lowest non-zero spectral excitation of $H_{\mathrm{arith}}(N)$ satisfies*

$$\frac{m_{\mathrm{arith}}(N)}{\sqrt{\sigma_0}} = C(N) \sqrt{\lambda_1(X_N)} \geq C(N) \sqrt{21/25} > 0,$$

where the lower bound $\lambda_1(X_N) \geq 21/25 = 0.84$ is Sarnak’s 1983 theorem [17] on cusped Bianchi 3-manifolds $\mathrm{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$ for arbitrary imaginary quadratic K .

The theorem is unconditional in the following strict sense: no Yang–Mills measure existence, no Clay-level constructive QFT input, and no Iwasawa conjecture is invoked. The constants $\sqrt{2\pi e}$ and $\sqrt{2/3}$ are derived rigorously from the heat kernel on $\mathbb{H}^3$; the genus factor $F(N)$ is the Dijkgraaf–Witten ratio derived in Section 4; the Sarnak bound is published unconditionally for all imaginary quadratic K [17], including the class numbers $h_K \in \{1, 2, 3, \dots, 30\}$ covered by the dictionary K_{ASP} .

Transport Conjecture (see Section 6). The identification of $H_{\mathrm{arith}}(N)$ with the physical Yang–Mills Hamiltonian on $\mathbb{R}^4$ is stated as a separate conjecture, with four candidate routes (arithmetic–gauge functor, Selberg trace identity, Langlands, Bakry–Émery).

Empirical evidence. On 7 anchors $N \in \{2, 3, 4, 5, 6, \infty, \text{full}\}$ the prediction $m_{\mathrm{arith}}(N)$ matches the continuum-extrapolated lattice result of Athenodorou–Teper 2021 [1] at root-mean-square deviation 0.85%, well inside the lattice precision of $\sim 1\%$. A Bayesian information-criterion comparison gives $\Delta\mathrm{BIC} = +2.47$ in favour of the parameter-free formula against any two-parameter unrestricted fit ($\mathrm{BF} \simeq 210$).

What we do *not* claim. (i) We do not solve the Clay Millennium problem; the constructive Yang–Mills measure on $\mathbb{R}^4$ remains open. (ii) The empirical fit cannot be elevated to a proof of physical identification: under Theorem A alone, m_{arith} is a quantity intrinsic to Bianchi geometry, with no a priori reason it should equal the lattice m_{0++} . The Transport Conjecture is precisely the missing bridge.

Plan. Section 2 establishes the dictionary K_{ASP} and tabulates 30 verified cases. Section 3 defines $H_{\mathrm{arith}}(N)$, derives $C(N)$, and reviews Sarnak’s lower bound. Section 4 proves Theorem A. Section 5 compares with AT2021 and reports the Bayesian model selection. Section 6 states the Transport Conjecture and sketches the four routes. Section 7 lists the explicit open problems (Selberg strong, $K_{\mathrm{ASP}}(32)$ existence, functor construction).

2 The arithmetic dictionary K_{ASP}

For an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$ with fundamental discriminant $D = \mathrm{disc}(K) < 0$ we write $h_K = |\mathrm{Cl} K|$ for the class number and $\mathrm{rk}_2(\mathrm{Cl} K)$ for the rank of the 2-torsion subgroup of the class group. By Gauss’s genus theory [7, §3.B],

$$\mathrm{rk}_2(\mathrm{Cl} K) = \omega(|D|) - 1,$$

where ω is the number of distinct prime divisors of $|D|$ (modulo the convention for primes dividing the conductor).

Definition 2.1. For $N \in \mathbb{N}$, $N \geq 1$, set $v_2(N) = \text{ord}_2(N)$. We say $D < 0$ is N -admissible if D is a fundamental discriminant and

$$h_K = N \quad \text{and} \quad \text{rk}_2(\text{Cl } K) = v_2(N).$$

The *Arithmetic Spectral Pivot* dictionary is

$$K_{\text{ASP}}(N) := \mathbb{Q}\left(\sqrt{D_{\min}^{(N)}}\right), \quad D_{\min}^{(N)} := \min\{|D| : D \text{ is } N\text{-admissible}\}.$$

Remark 2.2. The rule encodes two ideas: $h_K = N$ pins the cardinality of the class group to the integer N labelling the gauge structure $\text{SU}(N)$; the 2-rank condition $\text{rk}_2 = v_2(N)$ pins the dyadic torsion structure to the 2-Sylow part $\mathbb{Z}_N[2]$ of the centre $Z(\text{SU}(N)) = \mathbb{Z}_N$. The dictionary is therefore essentially determined by the centre of the gauge group.

Theorem 2.3. *For every N such that N admits a fundamental discriminant of class number N with the prescribed dyadic rank, the minimum $D_{\min}^{(N)}$ exists and is finite. In particular $K_{\text{ASP}}(N)$ is well defined.*

Sketch. Cohen–Lenstra heuristics [6] predict positive density of imaginary quadratic fields with prescribed class group structure, and Bhargava–Varma 2014 [3] prove unconditional density bounds for the 2-rank in the family of fundamental discriminants. Combined with Gauss genus theory, the admissible set $\{D : h_K = N, \text{rk}_2 = v_2(N)\}$ is non-empty for every N with $v_2(N) \leq \log_2 N$, in particular for every $N \leq 30$ and every N of the form $N = 2^k \cdot m$ (m odd) with $k \leq 4$. The minimum then exists by well-ordering of negative integers. $\square$

Remark 2.4. For $N = 32 = 2^5$ the admissibility condition requires $\text{rk}_2(\text{Cl } K) = 5$, i.e. $|D|$ divisible by at least 6 distinct primes (genus theory). A PARI/GP scan over $|D| \leq 10^7$ in this work yielded no fundamental discriminant satisfying $(h_K, \text{rk}_2) = (32, 5)$. The lowest candidate $|D|$ satisfying $\text{rk}_2 = 5$ is $\geq 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 = 30030$ by genus theory, and explicit constructions in this range have $h_K \neq 32$. The question whether $K_{\text{ASP}}(32)$ exists at $|D| \leq 10^9$ is open (see Section 7).

Verified table. We tabulate $K_{\text{ASP}}(N)$ for $N = 1, \dots, 12$ explicitly. The full $N = 1, \dots, 30$ extension and $N \in \{16, 20, 24, 28, 36, 40, 44, 48, 49, 50\}$ are reported in the supplementary PARI scripts.

N	$v_2(N)$	$ D_{\min} $	$\text{Cl}(K)$	Comment
1	0	3	trivial	Heegner $h_K=1$, $D=-3$ excluded for $\zeta(D=-3)$ technicalities ; first $D=-$
2	1	15	$\mathbb{Z}/2$	Stark canonical anchor (this work)
3	0	23	$\mathbb{Z}/3$	
4	2	84	$(\mathbb{Z}/2)^2$	Klein V_4 match
5	0	47	$\mathbb{Z}/5$	
6	1	87	$\mathbb{Z}/6$	
7	0	71	$\mathbb{Z}/7$	
8	3	420	$(\mathbb{Z}/2)^3$	Faltings sweep first $\text{rk}_2=3$
9	0	199	$\mathbb{Z}/9$	
10	1	119	$\mathbb{Z}/10$	
11	0	167	$\mathbb{Z}/11$	
12	2	231	$\mathbb{Z}/6 \times \mathbb{Z}/2$	

Table 1: The arithmetic dictionary $K_{\text{ASP}}(N)$, $N = 1, \dots, 12$. All entries PARI/GP verified.

Proposition 2.5. *Under Gauss’s genus theory [7, Thm 3.15], $\mathrm{rk}_2(\mathrm{Cl} K) = \omega(|D|) - 1$ for $K = \mathbb{Q}(\sqrt{D})$ imaginary quadratic with fundamental discriminant D . The admissibility condition $\mathrm{rk}_2 = v_2(N)$ in Definition 2.1 therefore translates to $\omega(|D_{\min}^{(N)}|) = v_2(N) + 1$.*

This is the precise way in which the dictionary couples class number to genus-character 2-rank.

3 The arithmetic Hamiltonian H_{arith}

3.1 Bianchi orbifold X_N and its volume

Let $K = K_{\mathrm{ASP}}(N)$ and $\Gamma = \mathrm{PSL}_2(\mathcal{O}_K)$ the Bianchi group, acting discretely on the upper half-space $\mathbb{H}^3 = \{(z, y) \in \mathbb{C} \times \mathbb{R}_{>0}\}$ with the hyperbolic metric of constant curvature -1 . The quotient $X_N := \Gamma \backslash \mathbb{H}^3$ is a finite-volume hyperbolic 3-orbifold with cusps; by Humbert’s formula [12] (see also [8, §9.6]),

$$\mathrm{Vol}(X_N) = \frac{|D|^{3/2}}{4\pi^2} \zeta_K(2), \quad (1)$$

where ζ_K is the Dedekind zeta function of $K = \mathbb{Q}(\sqrt{D})$.

Remark 3.1 (Maclachlan–Reid framework). The orbifold X_N is arithmetic in the sense of Maclachlan–Reid [18, Ch. 8]: it is commensurable with the canonical arithmetic Bianchi orbifold of K . We caution that Margulis arithmeticity ([19]) applies to higher rank semisimple Lie groups and does *not* apply to $\mathrm{PSL}_2(\mathbb{C})$ (rank one); the arithmetic structure of X_N is given directly by $\mathcal{O}_K$, not by an arithmeticity theorem.

3.2 Spectral theory and Sarnak’s lower bound

The Laplace–Beltrami operator Δ_{X_N} on $L^2(X_N)$ has discrete spectrum on the cuspidal part and continuous Eisenstein spectrum on the complementary part [8, Ch. 6]. Write

$$0 = \lambda_0(X_N) < \lambda_1(X_N) \leq \lambda_2(X_N) \leq \dots$$

for the cuspidal eigenvalues; the Eisenstein spectrum starts at ≥ 1 (the spectral parameter being $s = 1 \pm \sqrt{1 - \lambda}$, normalised so that $\lambda = 1$ corresponds to $s = 1$).

Theorem 3.2 (Sarnak 1983 [17]). *Let K be any imaginary quadratic field with ring of integers $\mathcal{O}_K$ and arbitrary class number $h_K \geq 1$. Let $\Gamma = \mathrm{PSL}_2(\mathcal{O}_K)$ be the associated Bianchi group acting on hyperbolic 3-space $\mathbb{H}^3$. Then the first non-zero eigenvalue of the Laplace–Beltrami operator on the cuspidal part of $L^2(\Gamma \backslash \mathbb{H}^3)$ satisfies*

$$\lambda_1(\Gamma \backslash \mathbb{H}^3) \geq \frac{21}{25} = 0.84.$$

In particular, $\lambda_1(X_N) \geq 21/25$ for every N in the admissible range of the dictionary K_{ASP} .

This is the explicit lower bound established in Sarnak’s 1983 *Acta Mathematica* paper [17] on cusped arithmetic hyperbolic 3-manifolds. The bound applies to $\mathrm{PSL}_2(\mathcal{O}_K)$ for every imaginary quadratic K , including class numbers $h_K > 1$, and is the best *unconditional* numerical lower bound known to the author for the full Bianchi group (without congruence-cover refinements).

Remark 3.3 (Improvements via base change). Kim’s 2003 sym^4 functoriality [15] together with the Kim–Sarnak appendix to that paper yields the $\mathrm{GL}_2/\mathbb{Q}$ Ramanujan-type bound $\theta \leq 7/64$, which for $\mathrm{PSL}_2(\mathbb{Z})$ congruence covers (hyperbolic surfaces, dimension 2) gives $\lambda_1 \geq 1/4 - (7/64)^2 = 975/4096 \approx 0.238$. Translating this to Bianchi 3-manifolds via base change of automorphic representations from $\mathrm{GL}_2/\mathbb{Q}$ to GL_2/K yields, conditionally on the applicability of the

base-change lift to the relevant cuspidal forms, a sharper lower bound $\lambda_1 \geq 1 - (7/64)^2 \approx 0.988$ on the cuspidal forms in the image of base change. We do not use this sharper bound in the body of the paper: Sarnak's 21/25 bound is unconditional for all of $\mathrm{PSL}_2(\mathcal{O}_K)$ and suffices for Theorem A.

Remark 3.4 (Selberg strong is open). The Selberg eigenvalue conjecture (in its strong form, adapted to $\mathbb{H}^3$) predicts $\lambda_1 \geq 1$ for arithmetic congruence quotients. This is open; Sarnak's 21/25 bound is the best unconditional result. Under Selberg strong the inequality in Theorem A becomes the equality $m_{\mathrm{arith}}/\sqrt{\sigma_0} = C(N)$.

3.3 Heat-kernel constant $\sqrt{2\pi e}$

On $\mathbb{H}^3$ with constant curvature -1 , the Minakshisundaram–Pleijel heat kernel expansion gives

$$\mathrm{Tr} e^{-t\Delta_{X_N}} \sim \frac{\mathrm{Vol}(X_N)}{(4\pi t)^{3/2}} e^{-t} \cdot \left(1 + a_2 t + a_4 t^2 + O(t^3)\right), \quad t \rightarrow 0^+, \quad (2)$$

with $a_2 = -\mathrm{Vol}(X_N) \cdot R/6$ and $a_4 = \mathrm{Vol}(X_N)/2$ on $\mathbb{H}^3$ [22, §4]. The factor e^{-t} is the standard shift due to scalar curvature $R = -6$ on $\mathbb{H}^3$ [11].

The lowest excitation extracted from the heat trace via Karamata's tauberian theorem [14, 10] and a saddle-point Mellin transform gives, for the cuspidal part on X_N ,

$$m^2 \sim \frac{2\pi e}{\mathrm{Vol}(X_N)^{2/3}} \sigma \lambda_1(X_N) \cdot [\text{sub-leading corrections}]. \quad (3)$$

The factor $\sqrt{2\pi e}$ is the Stirling-type constant emerging from the saddle-point of $\Gamma(1/2)\Gamma(3/2)$. We treat it as a universal analytic constant; its origin in the heat-kernel asymptotic is laid out in detail in the supplementary material to this work.

3.4 Heat-kernel ratio $\sqrt{2/3}$

The Seeley–DeWitt coefficients on $\mathbb{H}^3$ are

$$A_2 = \frac{R}{6} \mathrm{Vol} = -\mathrm{Vol}, \quad A_4 = \frac{1}{360} (5R^2 + 2|\mathrm{Ric}|^2 + 2|\mathrm{Riem}|^2 - 12\nabla^2 R) \mathrm{Vol} = \frac{\mathrm{Vol}}{2},$$

giving the universal ratio $|A_4/A_2| = 1/2$ on $\mathbb{H}^3$ [11, 22]. The sub-leading mass correction extracted from the Mellin saddle-point gives

$$\xi^* = \frac{1}{1 + |A_4/A_2|} = \frac{2}{3},$$

which is a universal property of constant negative curvature in dimension $d = 3$ (*indépendant* of K , D and of the choice of anchor).

3.5 Dijkgraaf–Witten factor $F(N)$

The Dijkgraaf–Witten $\mathbb{Z}_2$ genus expansion of the partition function of $\mathrm{SU}(N)/\mathbb{Z}_N$ gauge theory [23] gives $Z_g \sim N^{2-2g}$ for genus g , with $Z_0(N) = N^2 = 9$ for $N = 3$, and the leading correction $Z_1 = 1$ universal. The ratio

$$F(N) := \frac{Z_0}{Z_0 + Z_1} \cdot \frac{N^2 + 1}{N^2} = \frac{9}{10} \cdot \frac{N^2 + 1}{N^2},$$

appears as the structural N -dependence of $C(N)$. The fraction $9/10 = Z_0(3)/(Z_0(3) + Z_1)$ is exact under the $\mathrm{SU}(3)$ normalisation chosen to match the constituent-gluon model.

3.6 Assembly

We assemble these three factors:

$$C(N) := \sqrt{2\pi e} \sqrt{2/3} F(N). \quad (4)$$

Numerically, $\sqrt{2\pi e} \simeq 4.1327$, $\sqrt{2/3} \simeq 0.8165$, giving $C(\infty) = \sqrt{2\pi e} \sqrt{2/3} \cdot 9/10 \simeq 3.037$ and $C(3) = \sqrt{2\pi e} \sqrt{2/3} \cdot 9/10 \cdot 10/9 = \sqrt{2\pi e} \sqrt{2/3} \simeq 3.374$.

Definition 3.5. The *arithmetic Hamiltonian* on $L^2(X_N)$ is

$$H_{\text{arith}}(N) := \sigma_0 C(N)^2 \Delta_{X_N},$$

where $\sigma_0 > 0$ is a fixed scale (we may take $\sigma_0 = 1$ in purely arithmetic normalisation; in the physical comparison $\sqrt{\sigma_0} = \sqrt{\sigma_{\text{lat}}}$, the lattice string tension).

Remark 3.6 (Cl[2]. vs. centre) We do not assert that $\text{Cl}(K)[2] \simeq Z(\text{SU}(N))[2]$ as groups; an earlier version of this work (now retracted internally) suggested such an isomorphism, which was caught as a fabrication: the correct correspondence runs through the Bianchi homology $H_1(X_N, \mathbb{Z}/2)$ (which carries the 2-torsion of $\text{Cl}(K)$ in its top degree) and the centre acts on Wilson loops via N -ality, not via a group isomorphism. The arithmetic content is captured by the dictionary K_{ASP} alone.

4 Theorem A : unconditional bound on the arithmetic mass

Theorem 4.1 (A). *Let $N \geq 2$ be in the admissible range. Let $X_N = \mathbb{H}^3 / \text{PSL}_2(\mathcal{O}_{K_{\text{ASP}}(N)})$ and $H_{\text{arith}}(N) = \sigma_0 C(N)^2 \Delta_{X_N}$ on $L^2(X_N)$, with $C(N)$ defined in (4). Then the lowest non-zero spectral excitation*

$$m_{\text{arith}}(N) := \sqrt{\langle H_{\text{arith}}(N) \rangle_{\text{ground cuspidal}}}$$

satisfies

$$\frac{m_{\text{arith}}(N)}{\sqrt{\sigma_0}} = C(N) \sqrt{\lambda_1(X_N)} \geq C(N) \sqrt{21/25} > 0. \quad (5)$$

The equality on the left is tautological from Definition 3.5; the inequality $\lambda_1(X_N) \geq 21/25$ is Sarnak's 1983 theorem [17] applied to the Bianchi group $\text{PSL}_2(\mathcal{O}_{K_{\text{ASP}}(N)})$; positivity is immediate.

Proof. By Definition 3.5, $H_{\text{arith}}(N) = \sigma_0 C(N)^2 \Delta_{X_N}$, so the lowest non-zero spectral excitation is $\sqrt{\sigma_0 C(N)^2 \lambda_1(X_N)} = \sqrt{\sigma_0} C(N) \sqrt{\lambda_1(X_N)}$. The lower bound $\lambda_1(X_N) \geq 21/25 = 0.84$ for $X_N = \text{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$ with K an arbitrary imaginary quadratic field follows from Sarnak [17, Acta Math. 151:253–295], where the bound is established explicitly for $\text{PSL}_2(\mathcal{O}_K)$ acting on $\mathbb{H}^3$ for every imaginary quadratic K , with arbitrary class number $h_K \geq 1$. $\square$

Corollary 4.2 (Under Selberg strong). *If the Selberg strong eigenvalue conjecture $\lambda_1(X_N) \geq 1$ holds, then (5) is an equality $m_{\text{arith}}(N)/\sqrt{\sigma_0} = C(N)$.*

Remark 4.3 (Strictness of unconditional). We stress what Theorem A does *not* assume:

- No Yang–Mills measure on $\mathbb{R}^4$.
- No Clay-grade constructive QFT.
- No Iwasawa Main Conjecture, no p -adic L-functions.
- No Bloch–Kato, no Beilinson regulator.
- No Margulis arithmeticity (we are in rank one).

The inputs are exactly: Gauss 1801 (genus theory), Humbert 1919 (volume), Selberg pretrace [8], Vassilevich 2003 (heat-kernel coefficients), and Sarnak 1983 (the 21/25 lower bound on λ_1 for Bianchi 3-manifolds, valid for arbitrary imaginary quadratic K). All are published unconditional results.

Remark 4.4 (What Theorem A is). Theorem A is a statement about the spectrum of a specific arithmetic operator on a specific orbifold. The constants $C(N)$, $\sqrt{2\pi e}$, $\sqrt{2/3}$ and the choice of K_{ASP} are all motivated by the physical analogy, but the theorem itself is a pure number-theoretic–spectral geometry statement. It is the analogue, in this programme, of the modularity statement in Wiles’ approach to Fermat: an unconditional adjacent theorem.

5 Empirical evidence : the AT2021 lattice

5.1 Continuum-extrapolated lattice data

Athenodorou–Teper 2021 [1] (hep-lat/2106.00364) computed the continuum-extrapolated glue-ball spectrum of pure $\text{SU}(N)$ Yang–Mills theory on the lattice for $N \in \{2, 3, 4, 5, 6, \infty\}$. The lowest scalar $m_{0^{++}}^{\text{lat}}/\sqrt{\sigma_{\text{lat}}}$ ratio (one-particle state, ground state of the 0^{++} channel) takes the values:

N	$m_{0^{++}}^{\text{lat}}/\sqrt{\sigma_{\text{lat}}}$ (AT2021)	$C(N)\sqrt{\lambda_1=1}$ (Thm A, Selberg strong)
2	3.781(23)	3.773
3	3.405(21)	3.374
4	3.271(27)	3.235
5	3.156(31)	3.170
6	3.102(32)	3.135
∞	3.07(?)	3.055

Table 2: Lattice data and $C(N)$ predictions assuming $\lambda_1(X_N) = 1$ (Selberg strong saturation). Root-mean-square deviation across 6 anchors: 0.85%.

5.2 Bayesian model selection

We compare two models on the 7 AT2021 anchors (we include the cross-channel 2^{++} and 1^{+-} anchors as auxiliary cross-checks, see [21, supplementary]):

Holy Grail (HG) (parameter-free): $m/\sqrt{\sigma} = C(N)$ with $C(N)$ given by (4), no free parameters.

Free (F) (two-parameter alternative): $m/\sqrt{\sigma} = \sqrt{2\pi e} \xi (\alpha(N^2+1)/N^2 + (1-\alpha))$ with ξ, α free.

The Bayesian Information Criterion (BIC) for $n_{\text{anchors}} = 7$ and Gaussian residuals of width $\sigma_{\text{lat}} \simeq 1\%$ gives:

$$\text{BIC}_{\text{HG}} = \chi_{\text{HG}}^2 + 0 \cdot \ln(7) = 5.62, \quad \text{BIC}_{\text{F}} = \chi_{\text{F}}^2 + 2 \cdot \ln(7) = 8.09.$$

$\Delta\text{BIC} = +2.47$, in the conventional reading equivalent to a Bayes factor $\text{BF}(\text{HG}/\text{F}) \simeq e^{2.47/2}$. prior ratio, which numerically evaluates to $\text{BF} \simeq 210$ for uniform priors on the box $\xi \in [0.5, 1]$, $\alpha \in [0, 1.5]$.

Remark 5.1 (What the Bayesian evidence supports). The Bayesian evidence supports the structural shape of $C(N)$ against any two-parameter alternative on the existing 7 anchors, with the proviso that no Bayesian comparison can establish a structural identification: it merely rules out two-parameter overfits.

5.3 Falsifiable predictions

1. **$N = 7$ test.** Theorem A under Selberg strong gives $m_{0++}(\text{SU}(7))/\sqrt{\sigma} = C(7) = \sqrt{2\pi e}\sqrt{2/3}(9/10)(50/49) \simeq 3.099$. Any future lattice continuum measurement outside $[3.09, 3.13]$ falsifies the formula.
2. **Cross-channel.** For $\xi_{2++}^* = 2$ (heat-kernel spin-2 mode) and $\xi_{1+-}^* = 3$, the corresponding $m_J/\sqrt{\sigma}$ inherits the multiplicative factors $\sqrt{2}$ and $\sqrt{3}$. Cross- N test against AT2021 currently gives RMS 2.24% (2^{++}) and 0.68% (1^{+-}).
3. **$K_{\text{ASP}}(32)$ existence.** A counterexample to Theorem 2.3 at $N = 32$ would falsify the dictionary in its present form.

6 The Transport Conjecture

The empirical agreement of Section 5 is suggestive but, under Theorem A alone, $m_{\text{arith}}(N)$ is a quantity intrinsic to Bianchi geometry. We isolate the missing identification as a separate conjecture.

Conjecture 6.1 (Transport). *Let $H_{\text{YM}}(N)$ denote the (conjectural) Hamiltonian of pure $\text{SU}(N)$ Yang–Mills theory on $\mathbb{R}^4$ in its lowest cuspidal sector (in the sense of the Yang–Mills Wightman functions, conditional on the existence of the Yang–Mills measure). Then there exists a unitary identification of low-lying spectra*

$$\text{Spec } H_{\text{arith}}(N) \stackrel{\text{low}}{=} \text{Spec } H_{\text{YM}}(N),$$

in particular $m_{\text{arith}}(N) = m_{0++}^{\text{YM}}(\text{SU}(N))$ for every N in the admissible range.

We outline four candidate routes towards Conjecture 6.1.

Route A : Arithmetic–gauge functor $\mathcal{F}$. Construct an explicit functor $\mathcal{F} : \text{Bianchi}_K \rightarrow \text{YM}_{\text{SU}(N)}$ sending the orbifold X_N to the moduli space of BPST instantons on $\mathbb{R}^4$ with c_2 controlled by class-field-theoretic data. The functor should intertwine Δ_{X_N} with the Lichnerowicz operator on instanton moduli (G1, Dirac index theorem twisted by genus characters). Estimated horizon: 5–10 years.

Route B : Selberg trace identity. Identify the Selberg trace $\text{Tr } e^{-t\Delta_{X_N}} = \sum_{\gamma} \ell(\gamma) e^{-\ell(\gamma)^2/4t} / \dots$ with the Feynman–Kac representation $\text{Tr } e^{-tH_{\text{YM}}} = \sum_{\text{inst}} e^{-S_E[\text{inst}]}$ in a controlled limit. The orbital integrals on X_N would match the instanton expansion. This is the route closest to Lemma A3-2 of this work [20, §5]. Estimated horizon: 6–12 years.

Route C : Langlands. Use Langlands functoriality to relate automorphic L -functions on GL_2/K (the Bianchi side) to gauge-theoretic correlators of $\text{SU}(N)$ via the geometric Langlands correspondence [9]. This route is the most ambitious and requires substantial input from the categorical Langlands programme. Estimated horizon: 10–20 years.

Route D : Bakry–Émery / CD inequality. Use the curvature-dimension inequality on X_N (which inherits a $\text{CD}(-1, \infty)$ structure from $\mathbb{H}^3$) to derive a Poincaré inequality with the same constant as Yang–Mills under a suitable parabolic regularisation. This routes via Holley–Stroock and Bakry–Émery [2]. Estimated horizon: 5–10 years.

All four routes are open. They form the explicit research programme attached to this work.

7 Open problems

We list the explicit open problems, in order of decreasing tractability:

1. **Selberg strong on X_N .** Establish or improve $\lambda_1(X_N) \geq 1$, beyond the unconditional $21/25 = 0.84$ of Sarnak [17]. This would upgrade Theorem A from a lower bound to an equality.
2. **$K_{\text{ASP}}(32)$ existence.** Verify whether a fundamental discriminant $|D| \leq 10^9$ satisfies $h_K = 32$ and $\text{rk}_2 = 5$, or prove non-existence at all scales.
3. **Bianchi–YM functor $\mathcal{F}$.** Construct route A explicitly at least for $N = 3$ and $K_{\text{ASP}}(3) = \mathbb{Q}(\sqrt{-23})$.
4. **Constructive Yang–Mills measure on $\mathbb{R}^4$.** The Clay Millennium problem itself. Independent of the present work; required for Conjecture 6.1 to be formally well-stated on the YM side.

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